

WASP-10b

R-0727

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_221208 · created 2026-07-28T00:31:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459921.6673 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.06
Transit depth (Rp/Rs)^2	3.98 +/- 2.41 [%]
Orbital Inclination (inc)	85.87 +/- 1.86
Airmass coefficient 1 (a1)	2.23 +/- 1.49
Airmass coefficient 2 (a2)	-0.61 +/- 0.63
Scatter in the residuals of the lightcurve fit is	69.95 %
Best Comparison Star	#2 - [244, 307]
Optimal Aperture	3.0
Optimal Annulus	7.51
Transit Duration (day)	0.062 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2 ± 0.06 vs 0.1592 (deviation 0.68σ)
Duration fit vs published	0.062 d vs 0.0946 d (deviation 0.99σ)
Mid-time O–C	-12.0 min
Depth SNR	3.3
Residual scatter	69.95 %

Light curve

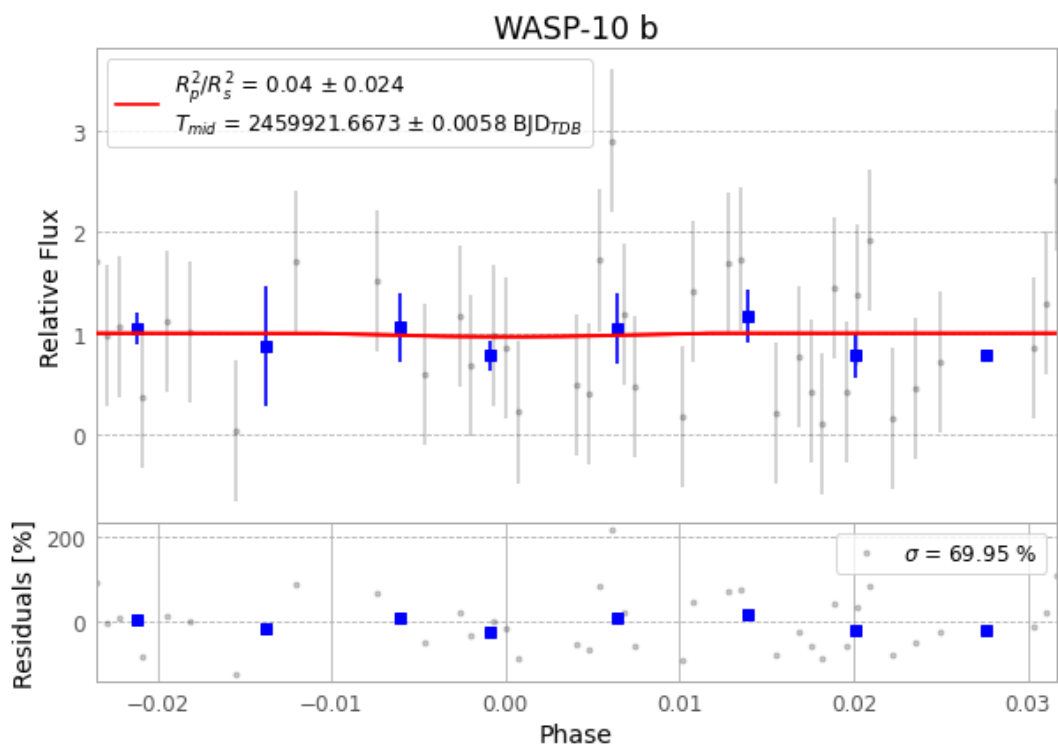


Figure 1 — FinalLightCurve_WASP-10 b_2022-12-07.png (SHA-256 7ddb431d719656eb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 147], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 134910b7de401356...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), WASP-10221208020915.FITS → WASP-10221208061815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-221208.log (7025cd8ecc84...), FinalLightCurve_WASP-10 b_2022-12-07.png (7ddb431d7196...), FinalLightCurve_WASP-10 b_2022-12-07.csv (679ce30f10fc...)

Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:31Z.